

2 Description

The STM32C091xB/xC STM32C092xB/xC mainstream microcontrollers are based on high-performance Arm® Cortex®-M0+ 32-bit RISC core operating at up to 48 MHz frequency. Offering a high level of integration, they are suitable for a wide range of applications in consumer, industrial and appliance domains and ready for the Internet of Things (IoT) solutions.

The devices incorporate a memory protection unit (MPU), high-speed embedded memories (up to 36 Kbytes of SRAM and up to 256 Kbytes of flash program memory with read and write protection, proprietary code protection, and securable area), DMA, an extensive range of system functions, enhanced I/Os, and peripherals. The devices offer standard communication interfaces (two I²Cs, two SPI / one I²S, one FDCAN, and four USARTs), one 12-bit ADC (2.5 MSps) with up to 19 channels, a low-power RTC, an advanced control PWM timer, five general-purpose 16-bit timers, a 32-bit general-purpose timer, two watchdog timers, and a SysTick timer.

The devices operate within ambient temperatures from -40 to 125°C and with supply voltages from 2.0 V to 3.6 V. Optimized dynamic consumption combined with power-saving modes allows the design of low-power applications.

The devices are housed in packages with 20 to 64 pins.

Table 2. STM32C091xB/xC and STM32C092xB/xC device features and peripheral counts

Peripheral		STM32C09x					
		_FB / _FC	_EC	_GB / _GC	_KB / _KC	_CB / _CC	_RB / _RC
Flash memory (Kbyte)		128/256	256	128/256			128/256 ⁽¹⁾
SRAM (Kbyte)		36 (STM32C091xx) / 30 (STM32C092xx)					
Timers	Advanced control	1 (16-bit)					
	General-purpose	1 (32-bit)					
		5 (16-bit)					
	SysTick	1					
	Watchdog	2					
Comm. interfaces	SPI [I2S] ⁽²⁾	2 [1] + 4 extra through USARTs					
	I2C	2					
	USART	4					
RTC		Yes					
FDCAN		No (STM32C091xx) / Yes (STM32C092xx)					
GPIOs (all 5V-tolerant)		18	22	26	30	45	61
DMA channels		7					
Wakeup pins		4	5	5	5	5	6
12-bit ADC channels (ext. + int.)		11 + 4	13 + 4	13 + 4	14 + 4	17 + 4	19 + 4
Max. CPU frequency		48 MHz					